

An Internship Report
Of
Python Development Virtual Internship

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech (Computer Science and Engineering)



By
Roll Number of Student: 25scs1003002320
Name of Student: HARSH
Batch: 2025-2029

CANDIDATE'S DECLARATION

I, HARSH, do hereby declare that the internship report titled “Python Development Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: HARSH

Roll No.: 25scs1003002320

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the upSkill Campus team and its industry partner UniConverge Technologies Pvt. Ltd. for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of Python Development. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:-

Student Name: HARSH

Roll No.: 25scs1003002320

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3. INTERNSHIP OFFER LETTER



C-56/11, Ground floor, Near IT Stellar Park, Sector-62, Noida, U.P. - 201309

Email: support@upskillcampus.com, Ph. No. 9354068856

Internship Offer Letter

Date: **15-07-2026**

Name: **Harsh**

Email: **harsh.25scs1003002320@iilm.edu**

Location: **Remote**

Dear **Harsh**,

We are pleased to inform you that **Upskill Campus** alongwith its industry Partner **UniConverge Technologies Pvt. Ltd.** has agreed to provide you internship as **Python Intern**. Your appointment of service is based on the following terms and conditions.

1. The internship shall start from **15-07-2026**
2. **Internship:** The internship period will be 4 to 6 weeks, starting from the date of joining.
3. **Working Hours:** The internship runs from 10:00 AM to 6:00 PM.

On behalf of both the Company, we wish you every success in your position and trust that our relationship will be long and mutually rewarding.

Sincerely

For Upskill Campus and UniConverge Technologies Pvt. Ltd.



Prabha Singh
HR Manager

4. INTERNSHIP COMPLETION CERTIFICATE

INTERNSHIP



CERTIFICATE OF COMPLETION

awarded to

Harsh

for successfully completing the Internship Program from **15-07-2026** to **29-08-2026**.

During this internship, the candidate actively participated in hands-on training and real industry project work in the domain of **Python**

This program was conducted in collaboration with **All India Council for Technical Education (AICTE)** and **UniConverge Technologies & upSkill Campus**.



Kaushlendra Singh Sisodia
Chief Mentor, The IoT Academy
Director, Uniconverge Technologies

Certificate ID: USC154635TIA

5. Project Description

5.1 Introduction

This report presents a detailed account of the Virtual Internship undertaken with Upskill Campus, in association with its industry partner UniConverge Technologies Pvt. Ltd., in the domain of Python Development. The internship commenced on 15 July 2026 and was carried out remotely as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida. The primary aim of this internship was to build a strong practical foundation in Python programming, file-system automation, and software development practices, applied through a real project titled the “Smart File Organizer.”

The internship followed a structured week-wise progression, beginning with an initial project plan and a basic file-scanning module, and advancing through file categorization, duplicate-file handling, summary statistics, logging, a menu-based user interface, testing, and documentation. The completed project, along with weekly progress reports, was submitted along with a testimonial and social-media sharing assignment as part of the final submission requirements of the programme.

5.2 Organization Profile

Upskill Campus is an online internship and skilling platform that partners with industry organizations to provide students with structured, hands-on internship experience. For this internship, Upskill Campus collaborated with its industry partner, UniConverge Technologies Pvt. Ltd., Noida, to offer the Python Development internship track. The internship was coordinated by Ms. Prabha Singh, HR Manager, and combined remote working hours with practical, project-based learning centered on Python automation.

5.3 Problem Statement

Manually organizing files that accumulate in a folder — documents, images, videos, audio, archives, and other file types — is a repetitive and error-prone task. Users often need to sort large numbers of files into meaningful categories while avoiding accidental overwriting of files that share the same name. The problem addressed by this internship project was to build a Python-based automation tool, the Smart File Organizer, that could scan a directory, classify files by extension, safely handle duplicate filenames, and provide clear feedback and logs on the outcome of the organization process.

5.4 Project Objectives

- To design and develop a Python-based file-scanning and categorization module.

- To implement automatic creation of category folders (Documents, Images, Videos, Audio, Archives, Others).
- To implement safe file-moving logic that prevents accidental overwriting of duplicate filenames.
- To add exception handling for file-access and permission-related errors.
- To implement summary statistics that report the number of files organized per category.
- To add operation logging for successfully organized and skipped files.
- To build a simple, menu-based user interface for selecting the directory to organize.
- To test the application across varied directory structures and file types.
- To prepare project documentation and a testimonial submission summarizing the internship experience.

5.5 Scope of the Project

The scope of this internship covers the complete development lifecycle of the Smart File Organizer — from an initial file-scanning module through categorization, duplicate handling, error handling, summary reporting, logging, a menu-driven interface, testing, and documentation. The project was developed and tested locally as a Python desktop automation script and was not deployed to any production or organizational system. The internship was conducted remotely over a period of 4 to 6 weeks under the guidance of Upskill Campus and UniConverge Technologies Pvt. Ltd.

5.6 Technologies and Tools Used

- Programming Language: Python
- Core Libraries: os, shutil, pathlib, logging
- Concepts Applied: file and directory management, exception handling, duplicate-file handling, modular programming
- Version Control & Hosting: Git and GitHub (project repository: upskillcampus)
- Development Approach: iterative, week-wise feature development with continuous testing and debugging

5.7 System Architecture

The Smart File Organizer follows a layered, modular design built up over the course of the internship:

- Scanning Layer: Scans the selected directory and identifies all files present.

- Categorization Layer: Maps file extensions to categories such as Documents, Images, Videos, Audio, Archives, and Others.
- Folder Management Layer: Automatically creates category folders as required.
- File-Moving & Duplicate-Handling Layer: Moves files into their category folders and assigns modified names to duplicates so existing files are never overwritten.
- Error-Handling Layer: Catches and manages file-access, permission, and path-related exceptions without stopping the overall process.
- Reporting Layer: Generates summary statistics and operation logs for successful and skipped files.
- Interface Layer: A simple menu-based interface through which the user selects the directory to organize.

5.8 Methodology

The internship followed a structured, week-wise methodology combining hands-on development, testing, and documentation, as summarized in the table below.

Week	Module	Description
1	Project Planning & File Scanning	Prepared the initial project plan for the Smart File Organizer and developed the basic file-scanning module.
2	Categorization & Duplicate Handling	Implemented file categorization by extension, automatic creation of category folders, file-moving logic, and duplicate-filename handling; tested with sample directories.
3	Summary Statistics & Reliability	Added summary statistics for organized files, improved exception handling and safety checks, and carried out extensive testing across file types and directory structures.
4	Interface, Logging & Documentation	Added a menu-based user interface, implemented operation logging, refined the summary report, optimized the code, fixed bugs, and began preparing project documentation.
Final	Testimonial & Submission	Completed the Smart File Organizer project, published it to GitHub, and prepared the video-testimonial and social-media sharing assignment as part of the final submission.

Note: The internship offer specified a duration of 4 to 6 weeks; the final week(s) were used to complete testing, documentation, and the testimonial submission described above.

5.9 Expected Outcomes

- Practical experience in Python file and directory handling using `os`, `shutil`, and `pathlib`.

- Ability to design and implement a rule-based file-categorization system.
- Skill in safely handling duplicate files without overwriting existing data.
- Understanding of exception handling for file-access and permission errors.
- Experience implementing logging and summary reporting in a Python application.
- Ability to design a simple, usable menu-driven interface for a command-line tool.
- Improved debugging, testing, and problem-solving skills through iterative development.
- A completed, documented project (Smart File Organizer) published to a public GitHub repository.

5.10 Certificates and Communication Proof

The Internship Offer Letter issued by Upskill Campus and UniConverge Technologies Pvt. Ltd., confirming the internship as Python Intern, is attached in Chapter 3 of this report. The completed project has been published to GitHub as further proof of work, and a video testimonial and social-media sharing assignment were prepared as part of the final submission requirements.

GitHub Project Repository: <https://github.com/harsh25scs1003002320-lab/upskillcampus>

- Tagged Organizations: Upskill Campus & UniConverge Technologies (P) Ltd.
- Hashtags: #upSkillCampus #UniconvergeTechnologies #SummerInternship2026 #PythonDevelopment #Python #SmartFileOrganizer #BTechCSE #IILMUniversity

6. Bibliography/References

- Upskill Campus & UniConverge Technologies Pvt. Ltd., “Internship Offer Letter,” dated 15-07-2026.
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- Python Software Foundation, logging — Logging facility for Python, <https://docs.python.org/3/library/logging.html>
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